Runan He

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I am an applied mathematician who models, analyzes and simulates multi-physics devices and nonlinear waves: electrostatic micro-electro-mechanical systems (MEMS), electrical discharge in their gas gaps, and nonlinear surface plasmons. My record: five refereed articles (*J. Differential Equations*, *Proc. R. Soc. A*, *Math. Methods Appl. Sci.*, *Discrete Contin. Dyn. Syst.*, *J. Math. Anal. Appl.*) and five preprints, two of them single-authored, six of the ten with numerical computations, including a positivity-preserving finite element solver for discharge; a short communication at ICM 2026; and six master's-level courses taught in Halle after four years of tutoring at Heriot-Watt. Next, I will make simulation and physics-informed machine learning trustworthy near device failure.

Employment

10/2024–present **Postdoctoral Researcher** (CSIC contract), PDE group, Instituto de Ciencias Matemáticas (ICMAT), Madrid, Spain

- Full time since 28 October 2024, planned until 28 October 2027; contract for research on singularities in the Navier–Stokes equations; no teaching duties.
- Alongside the contract research I wrote five preprints on MEMS and nonlinear surface plasmons (items 6–10), including my first paper without a former supervisor or mentor (item 6) and two single-author papers on electrical discharge (items 9 and 10).

10/2021–09/2024 **Postdoctoral Researcher**, applied analysis group, Martin-Luther-Universität Halle-Wittenberg, Halle (Saale), Germany

- Mentor: Prof. Tomas Dohnal. University-funded postdoctoral project *Bifurcation and Asymptotics in Maxwell Equations*, outside my thesis area (items 1 and 8).
- Taught six master's-level courses, including my lecture course *Evolution Equations*; began mentoring a PhD student in 2023.

09/2017–09/2021 **Tutorial Assistant**, Heriot-Watt University, Edinburgh, UK

09/2014–06/2017 **Teaching and Research Assistant**, Shandong University, Jinan, China

07/2013–12/2015 **Computer Programmer**, Shandong University, Jinan, China

- Created and managed the development schedule of a software project involving about 20 people.

Education

2017–2021 **PhD in Mathematics**, Heriot-Watt University, Edinburgh, UK

- Thesis: *Analysis of coupled PDE systems modelling micro-electro-mechanical systems* (September 2021); degree awarded 31 October 2021.
- Supervisors: Prof. Andrew Lacey and Dr. (now Prof.) Heiko Gimperlein. Internal examiner: Prof. Dominic Breit.

2014–2017 **MSc in Pure Mathematics**, Shandong University, Jinan, China

- Thesis: *Morse Theory and its Application to Semilinear Elliptic Equation*.

2013–2015 **MEng in Software Engineering**, Shandong University, Jinan, China

- Thesis: *The Design and Implementation of a Reliable Transport System Based on UDP for Large-Scale Data Transmission*.

2009–2013 **BSc in Mathematics and Applied Mathematics**, Hunan University, Changsha, China

Research interests

Models of physical and engineering systems: electrostatic MEMS (gas flow in the gap coupled to elastic plates and membranes); electrical discharge (drift-diffusion–Poisson systems); grain-boundary grooving in sintering (Mullins equation).

Nonlinear dynamics and mathematical physics: touchdown (quenching) singularities and self-similar solutions; stability of steady states and spectral thresholds (sparking voltage) in discharge models; bifurcation of nonlinear waves from eigenvalues of operator pencils, including PT-symmetric problems; nonlinear Maxwell equations.

Computational methods: positivity-preserving finite element methods; finite difference methods; Newton solvers with parameter continuation; numerical experiments near singularities.

Analytical methods: semigroup theory; fixed-point, compactness and monotone-operator methods; spectral and bifurcation theory; formal and rigorous asymptotics.

2027–2032 **Research plan:** *Trustworthy simulation and learning near device failure* (Lines 3 and 4 are new directions).

- Line 1. Structure-preserving simulation of multi-physics devices (first: a convergent positivity-preserving scheme for the transient model of item 7, building on the simulations of item 9).
- Line 2. Singularities and device failure: proofs and computational discovery (first: prove the stability of the constant self-similar profile conjectured in item 5).
- Line 3. Physics-informed machine learning with guarantees (first: a neural network that predicts the minimum gap and the pull-in voltage for the models of items 2–4).
- Line 4. Inverse problems, control and uncertainty quantification (first: identify permittivity, stiffness and damping from capacitance and pull-in data).

Publications

Authors in alphabetical order, as is customary in mathematics.

Refereed journal articles

1 T. Dohnal and **R. He**. Bifurcation and asymptotics of cubically nonlinear transverse magnetic surface plasmon polaritons. *Journal of Mathematical Analysis and Applications* 538(2) (2024), 128422. https://doi.org/10.1016/j.jmaa.2024.128422; arXiv:2311.17838 (with numerical computations)

2 H. Gimperlein, **R. He** and A. A. Lacey. Existence and uniqueness for a coupled parabolic–hyperbolic model of MEMS. *Mathematical Methods in the Applied Sciences* 47(7) (2024), 6310–6353. https://doi.org/10.1002/mma.9922; arXiv:2401.04128

3 H. Gimperlein, **R. He** and A. A. Lacey. Wellposedness of a nonlinear parabolic–dispersive coupled system modelling MEMS. *Journal of Differential Equations* 384 (2024), 193–251. https://doi.org/10.1016/j.jde.2023.11.028; arXiv:2312.00807

4 H. Gimperlein, **R. He** and A. A. Lacey. Wellposedness of an elliptic–dispersive coupled system for MEMS. *Discrete and Continuous Dynamical Systems* 43(9) (2023), 3485–3511. https://doi.org/10.3934/dcds.2023055; arXiv:2304.14257

5 H. Gimperlein, **R. He** and A. A. Lacey. Quenching for a semi-linear wave equation for micro-electro-mechanical systems. *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences* 478(2267) (2022), 20220490. https://doi.org/10.1098/rspa.2022.0490; arXiv:2210.14821 (with numerical computations)

Preprints

6 **R. He** and W. Xie. Analysis and numerics of a stationary drift-diffusion model for electrical discharge in MEMS. Submitted (2026). arXiv:2606.28223 (with numerical computations)

- My first paper without a former supervisor or mentor; W. Xie works at the company Feishu (Shanghai).

7 H. Gimperlein, **R. He** and A. A. Lacey. Analysis of a model for electrical discharge in MEMS. Submitted (2026). arXiv:2602.21439

8 T. Dohnal, M. Hanisch and **R. He**. Polychromatic localized waves with complex frequencies in nonlinear Maxwell equations with material dispersion. Submitted (2025). arXiv:2511.21467 (with numerical computations by M. Hanisch)

9 **R. He**. Global existence for a drift–diffusion model of electrical discharge in MEMS. Preprint (2026). (with numerical computations)
- Single author. Settles the global-existence question left open in item 7 for equal diffusivities and a bounded vacuum field, and proves a continuation criterion for the general case.

10 **R. He**. Asymptotic stability of stationary solutions of a drift–diffusion system for electrical discharge in MEMS micro-gaps. Preprint (2026). (with numerical computations)
- Single author. Explicit stability criterion for steady states and the sharp threshold (sparking) voltage between parallel electrodes.

In preparation

H. Gimperlein, **R. He** and A. A. Lacey. Self-similar solution to the Mullins equation modelling sintering.
- Result: existence of self-similar grain-boundary groove profiles for small contact angles.

H. Gimperlein, **R. He**, A. A. Lacey and B. Schörkhuber. Stable quenching for semilinear wave equations.
- Aim: to prove the stability, conjectured in item 5, of the constant self-similar quenching solution.

Doctoral thesis

2021 **R. He**. *Analysis of coupled PDE systems modelling micro-electro-mechanical systems*. PhD thesis, Heriot-Watt University, Edinburgh, September 2021. https://www.ros.hw.ac.uk/handle/10399/4629

Impact and follow-up work

2024 C. Walker. On a quasilinear parabolic–hyperbolic system arising in MEMS modeling. *Annali di Matematica Pura ed Applicata* 203(6) (2024), 2769–2783. https://doi.org/10.1007/s10231-024-01465-9
- Extends item 2 from one to two space dimensions, in a framework that also covers plate equations.

Computational experience

2026 **Discharge simulations and spectra** (items 9 and 10): time-dependent simulations of a discharge in a micrometre gap with the Townsend constants and mobilities of air; one-dimensional Scharfetter–Gummel computations of steady-state branches and of the spectrum of the linearization, checked under mesh refinement.

2026 **Finite elements** (item 6, with W. Xie): a solver for a stationary drift-diffusion model of electrical discharge in MEMS that keeps the carrier densities positive by construction, applied to two-dimensional domains and a three-dimensional axisymmetric geometry.

2024 **Numerical continuation** (item 1, with T. Dohnal): a finite difference Newton solver with parameter continuation computes bifurcation curves of nonlinear surface plasmons for two-layer examples and confirms the approximately quadratic error decay of the frequency asymptotics.

2022 **Numerical experiments** (item 5, with H. Gimperlein and A. A. Lacey) that support our conjecture on touchdown in a semilinear wave equation for MEMS.

2013–2015 **Software engineering:** MEng in Software Engineering (2015), with a thesis on designing and implementing a reliable UDP-based transport system; work as a computer programmer (see Employment).

Research funding and projects

My ICMAT position is associated with the first two awards below.

09/2024–08/2027 **Dynamics of singular incompressible fluids**, research project PID2023-152878NB-I00 (EUR 70,000), Spanish State Research Agency (AEI)

- PI: D. Córdoba (ICMAT). My role: postdoctoral researcher in the project team.

06/2024–05/2028 **ICMAT Severo Ochoa program**, institutional award CEX2023-001347-S, Spanish State Research Agency (AEI)

- PI: D. Córdoba (ICMAT). My role: postdoctoral researcher within the program.

10/2021–09/2024 **University-funded postdoctoral position**, Martin-Luther-Universität Halle-Wittenberg, Germany, for my postdoctoral project *Bifurcation and Asymptotics in Maxwell Equations*

Teaching experience

Teaching at three universities in three countries, in English, German and Chinese (2014–2024).

2021–2024 **Martin-Luther-Universität Halle-Wittenberg**, Halle (Saale), Germany: six one-semester master's-level courses, two contact hours per week each

- Lecturer, *Evolution Equations* (lecture course, English); I prepared some lecture notes and exercise sheets of my own, available on request.
- Lecturer, *Fachseminar* (subject seminar, German).
- Exercise classes in *Partial Differential Equations*, *Dynamical Systems* and *Nonlinear Analysis* (English) and *Mathematical Physics* (German).

2017–2021 **Heriot-Watt University**, Edinburgh, UK: tutorial assistant, on average five tutorial hours per week during semesters; undergraduate and some graduate level (English)

- Eight courses: *Calculus*, *Mathematics for Scientists and Engineers*, *Numerical Analysis*, *Mathematical Experiment*, *Partial Differential Equations*, *Real Analysis*, *Complex Analysis*, *Functional Analysis*.
- Training for tutorial assistants.

2014–2017 **Shandong University**, Jinan, China: teaching and research assistant in the undergraduate courses *Calculus* and *Partial Differential Equations* (Chinese)

Supervision and mentoring

Since 2023 I have mentored one doctoral student in joint research (below); I have not yet formally supervised a degree project or doctoral thesis.

2023–present **Mentoring of Maximilian Hanisch**, PhD student in T. Dohnal's group in Halle (Dohnal is his only formal supervisor; completion expected in 2027)

- A former student in my *Evolution Equations* course. Part of his doctoral project builds on item 1; in our joint preprint (item 8) he carried out the numerical computations.

Talks and posters

Conferences and workshops

07/2026 **Short communication.** International Congress of Mathematicians (ICM 2026), Philadelphia, USA. "Wellposedness of Nonlinear Coupled PDE Systems Modelling MEMS"

10/2025 **Poster.** Wave Propagation in Guiding Structures, Marseille, France. "Polychromatic Surface Plasmons with Complex Frequencies in Nonlinear Media"

07/2024 **Contributed talk.** 16th International Conference on Mathematical and Numerical Aspects of Wave Propagation (WAVES 2024), Berlin, Germany. “Monochromatic and Polychromatic Waves”

03/2024 **Contributed talk.** 94th Annual Meeting of GAMM, Magdeburg, Germany. “Bifurcation and Asymptotics in Nonlinear Surface Plasmon Polaritons”

07/2022 **Invited talk.** Workshop on Systems Theory and PDEs, Freiberg, Germany. “Wellposedness and Singularities in Nonlinear Coupled PDE Systems for MEMS”

03/2022 **Invited talk.** Conference on Mathematics of Wave Phenomena 2022, Karlsruhe, Germany (online). “Analysis of Nonlinear Coupled PDE Systems for MEMS”

Seminars and internal workshops

11/2024 **Invited talk** (workshop at my own institution). Research exposition workshop, ICMAT, Madrid, Spain. “Wellposedness and singularity formation for PDEs”

05/2024 **Seminar talk.** University of Bonn, Germany. “Wellposedness and singularity formation for PDE systems from natural science”

12/2023 **Invited seminar talk.** PDE seminar, TU Clausthal, Germany. “Bifurcation and Asymptotics in Maxwell Equations”

10/2021 **Invited seminar talk** (my own institution at the time). Oberseminar (research seminar), Martin-Luther-Universität Halle-Wittenberg, Germany. “Analysis for MEMS models”

2018–2020 **Seminar talks.** Postgraduate seminars of Heriot-Watt University and the Maxwell Institute, Edinburgh, UK

Organization, service and outreach

2021–2024 Oberseminar (research seminar), Martin-Luther-Universität Halle-Wittenberg: organized several sessions.

2018–2019 Assistant at Heriot-Watt University’s public Masterclass lectures for Edinburgh high-school students.

2016, 2017 Marker for the Chinese national university entrance examination (Gaokao).

2014–2017 Shandong University: organized seminars on the calculus of variations and PDEs.

Languages

Chinese (native); English (working language of my research and most of my teaching); German (about A2, self-assessed; I taught two courses in German).

Awards and scholarships

2015 Third Ranking Graduate Academic Scholarship, Shandong University, China.

2014 Exceptional Graduate Academic Scholarship, Shandong University, China.

2012 Third Prize, Mathematics Competition, Hunan Province, China.

2010 Second Prize, Mathematics Competition, Hunan Province, China.

References

Prof. Tomas Dohnal, postdoctoral mentor · tomas.dohnal@mathematik.uni-halle.de
Martin-Luther-Universität Halle-Wittenberg, Halle (Saale), Germany

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Heriot-Watt University, Edinburgh, UK

Prof. Heiko Gimperlein, doctoral supervisor · Heiko.Gimperlein@uibk.ac.at
Leopold-Franzens-Universität Innsbruck, Innsbruck, Austria

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